TAMPA INTERNATIONAL (TPA) DEPARTURES 2020–2025 AMERICAN · DELTA · SOUTHWEST
277,000 DEPARTURES

Three lies your flight stats are telling you

The industry loves a single average delay. At TPA, that average hides three different stories in the same numbers.

By **Naga Pavan Sathvik Reddy Sirigiri** · ISM 6419: Data Visualization for Storytelling ·

Data: U.S. Bureau of Transportation Statistics · ▶ Watch the walkthrough (REPLACE_WITH_YOUR_VIDEO_URL)

Every airline at Tampa International advertises the same thing: we run on time. American, Delta, and Southwest each report an average departure delay so small it looks reassuring, **11.2 minutes** .

But averages hide more than they show. Six years of flight records, 277,000 departures from TPA, reveal three different stories the airlines would rather you not notice. Each one can double or halve the delay you actually face.

Where you fly matters less than you think. When you fly matters more.

11.2 min

MEAN DEPARTURE DELAY · ALL CARRIERS

The *earlier* you fly, the more likely you are to leave on time.

A flight scheduled for 6 AM at Tampa leaves on time 83% of the time. A flight scheduled for 8 PM leaves on time only 48%, your odds of leaving on time nearly cut in half across one day. All three carriers show the same pattern.

Why does the day get worse? Because delays carry forward. Look at the right chart. The orange band, flights that were late because the same plane was already late from its previous trip, grows from 14% of all delay minutes at 6 AM to 69% by 8 PM. One late morning flight breaks the next five.

> *"It's not the airline you pick. It's the hour you pick."*

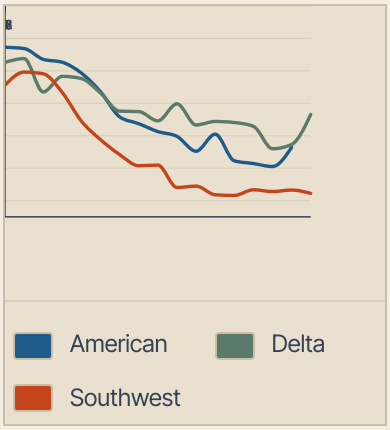
WHAT THIS MEANS FOR YOU

- Booking before **9 AM**? You're in the safe zone — on-time rates above 75% across all carriers.
- Stuck with an evening flight? Expect **roughly 1 in 2** chance of leaving late — pad your connection.
- The airline matters less than the hour. **Don't pay more** just to switch carriers at the same departure time.

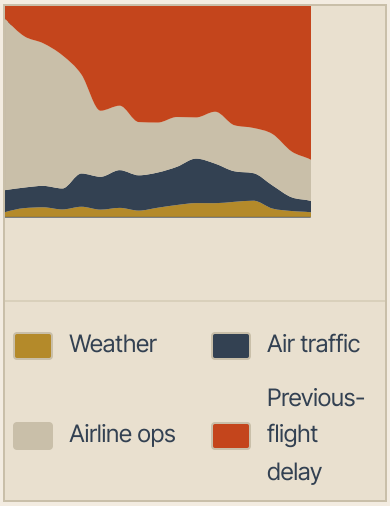
TAKEAWAY

A morning flight is on time nearly twice as often as an evening flight on the same airline. The reason is the cascade, delays compound across each plane's 5–6 daily legs. Book early.

WHAT: ON-TIME % BY HOUR



WHY: CAUSE COMPOSITION



Scheduled hour (local)

83%

ON-TIME AT 6 AM

48%

ON-TIME AT 8 PM

69%

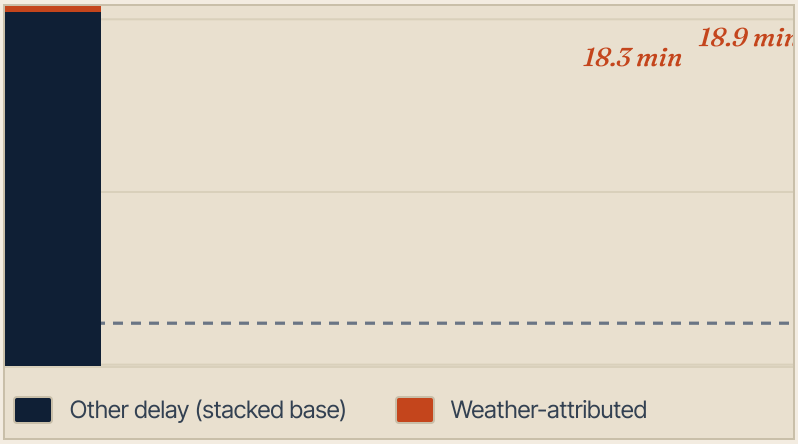
EVENING DELAY INHERITED

July is the worst month to fly Tampa. Not December.

Most travelers brace for Christmas-week delays. They're bracing for the wrong month.

December's average delay at Tampa is 11.2 minutes, right at the yearly average. July's is 19 minutes, and June is barely behind. The summer months are roughly three times worse than the fall.

The reason is sitting overhead. Tampa is on the Gulf Coast, where summer afternoons bring daily thunderstorms. Weather-attributed delay in summer is 9 times higher than in fall. The storms slow the morning down, and the cascade from earlier does the rest.



> *"Summer at Tampa isn't a vacation. It's a delay event tourists walk into."*

TAKEAWAY

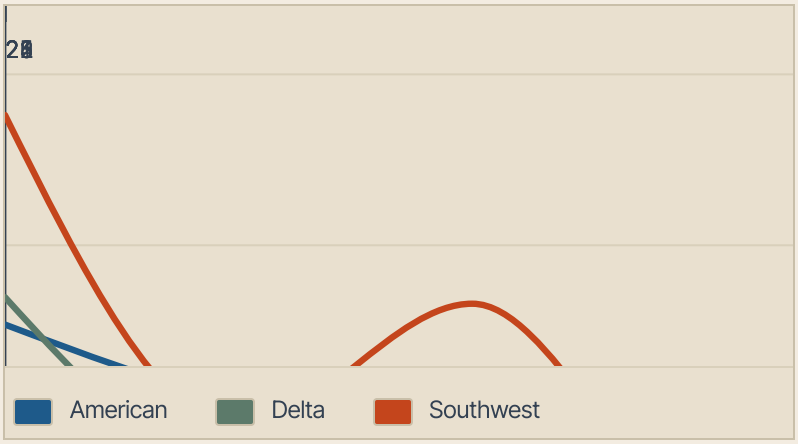
If your trip is flexible, fly in October or November. The average delay is one-third of July's. December is fine. Only the days right around New Year's spike.

Airlines *padded their schedules* in 2020 to look on time. Now they've cut it back.

Here's the part nobody tells you. A flight is "on time" if it arrives within the airline's scheduled flight time, but the airline picks that number.

In 2020, with empty skies during COVID, all three Tampa airlines built generous cushions into their schedules. American padded by **0 minutes**. Delta by **0 minutes**. Southwest by **0 minutes**. The plane only needed two hours, but the schedule said two-and-a-half. That made on-time stats look great.

By 2025, all three airlines had quietly cut the padding to about **0 minutes**. The aircraft didn't get faster. The schedules just got honest.



> *"The schedule isn't a prediction. It's a number every airline can move."*

WHY THIS MATTERS

A 2025 on-time rate that looks worse than 2020's might just mean more honest schedules, not worse operations. Year-over-year comparisons are apples-to-oranges across the whole industry.

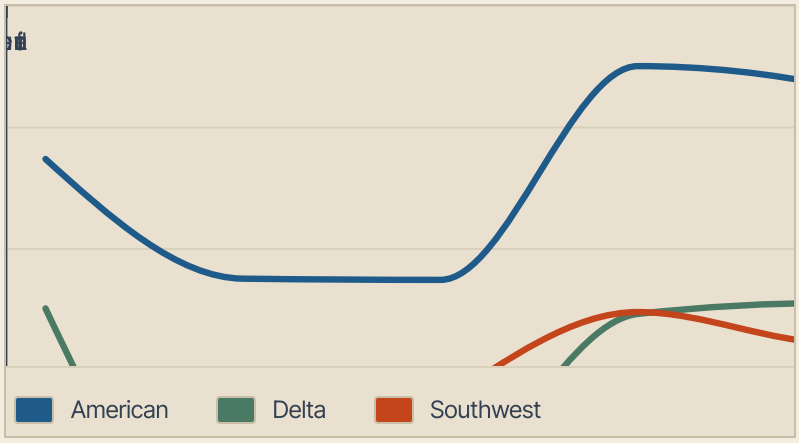
POSTSCRIPT ----- WHAT THE DATA HAS BEEN QUIETLY SAYING

Tuesday is the best day to fly Tampa, at every airline.

One more pattern hides in plain sight, and it's the easiest to act on.

A Tampa flight on Tuesday averages **0 minutes** of delay. The same flight on Sunday averages **0 minutes**. Tuesday is the best day at every airline in the dataset. American, Delta, and Southwest all agree.

But the real punch is what happens when the patterns stack. A Tuesday 6 AM flight averages **0 minutes**. A Sunday 8 PM flight averages **0 minutes**. That is a **0×** difference. The lies don't add together. They multiply.



> *"When the lies stack, they don't add. They multiply."*

If you can't change your hour or your month, change your day. Tuesday is the cheapest schedule decision a traveler can make. Three lies in one statistic, and when you know what to look for, the compounding shows everywhere.

THE COMPOUND

Tuesday 6 AM	0 min
Sunday 8 PM	0 min

0× *difference — same airport, same airlines, same passengers.*

So how late will your flight be?

It depends on three things the average hides: **what hour you fly**, **what month you fly**, and **how much cushion the airline put in your schedule**. Any one of them can double or halve your wait. Together, they explain almost everything.

The airlines aren't lying. They're just averaging, and a single number across **277,000 flights** can never answer one passenger's question. *Averages smooth. Stories specify.*

THREE LIES IN ONE STATISTIC

1 The 6 AM flight is on time **83%** of the time. The 8 PM flight, **48%**. The average tells you neither.

2 July's average delay is **3×** October's. The yearly average papers over an entire bad summer.

3 Airlines cut **6–12 minutes** of padding from their schedules since 2020. The same on-time rate means something different today than it did then.

And one more pattern the data quietly revealed. Keep scrolling. ↓

Source: U.S. DOT Bureau of Transportation Statistics, TPA departure detail files (American, Delta, Southwest), 2020–2025. Rows with invalid dates or missing delay / block times dropped. Cause shares use attributed delay minutes among positive-delay departures only. Padding chart: mean (scheduled - actual) block time by calendar year and carrier, all destinations.